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import java.util.InputMismatchException;
import java.util.Scanner;

public class Main {

    private static final Scanner scanner = new Scanner(System.in);
    private static final Library library = new Library();

    public static void main(String[] args) {
        System.out.println("
        System.out.println("
        System.out.println("
        LIBRARY MANAGEMENT SYSTEM  v1.0
        ");

        boolean running = true;

        while (running) {
            printMenu();
            int choice = readInt(" Enter your choice: ");

            switch (choice) {
                case 1:
                    handleAddBook();
                    break;
                case 2:
                    library.viewAllBooks();
                    break;
                case 3:
                    handleSearchBook();
                    break;
                case 4:
                    handleRemoveBook();
                    break;
                case 5:
                    handleAddUser();
                    break;
                case 6:
                    library.viewAllUsers();
                    break;
                case 7:
                    handleIssueBook();
                    break;
                case 8:
                    handleReturnBook();
                    break;
                case 9:
                    System.out.println("\n Thank you for using the Library
System. Goodbye!");
                    running = false;
                    break;
                default:
                    System.out.println(" [ERROR] Invalid choice. Please enter a
number between 1 and 9.");
            }

            System.out.println();
        }

        scanner.close();
    }

    private static void printMenu() {
        System.out.println("
        System.out.println("
        System.out.println("
        MAIN MENU
        ");
    }
}

```

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        System.out.println("    1. Add Book");
        System.out.println("    2. View All Books");
        System.out.println("    3. Search Book by Title");
        System.out.println("    4. Remove Book");
        System.out.println("    5. Add User");
        System.out.println("    6. View All Users");
        System.out.println("    7. Issue Book to User");
        System.out.println("    8. Return Book");
        System.out.println("    9. Exit");
        System.out.println("    ");
    }

    private static void handleAddBook() {
        System.out.println("\n --- ADD NEW BOOK ---");
        int id = readInt(" Enter Book ID : ");
        String title = readString(" Enter Book Title : ");
        String author = readString(" Enter Author Name : ");
        library.addBook(id, title, author);
    }

    private static void handleSearchBook() {
        System.out.println("\n --- SEARCH BOOK ---");
        String keyword = readString(" Enter title keyword: ");
        library.searchBookByTitle(keyword);
    }

    private static void handleRemoveBook() {
        System.out.println("\n --- REMOVE BOOK ---");
        int id = readInt(" Enter Book ID to remove: ");
        library.removeBook(id);
    }

    private static void handleAddUser() {
        System.out.println("\n --- ADD NEW USER ---");
        int id = readInt(" Enter User ID : ");
        String name = readString(" Enter User Name: ");
        library.addUser(id, name);
    }

    private static void handleIssueBook() {
        System.out.println("\n --- ISSUE BOOK ---");
        int bookId = readInt(" Enter Book ID: ");
        int userId = readInt(" Enter User ID: ");
        library.issueBook(bookId, userId);
    }

    private static void handleReturnBook() {
        System.out.println("\n --- RETURN BOOK ---");
        int bookId = readInt(" Enter Book ID: ");
        int userId = readInt(" Enter User ID: ");
        library.returnBook(bookId, userId);
    }

    private static int readInt(String prompt) {
        while (true) {
            System.out.print(prompt);
            try {
                int value = scanner.nextInt();
                scanner.nextLine();
                return value;
            } catch (InputMismatchException e) {
                System.out.println(" [ERROR] Please enter a valid integer.");
                scanner.nextLine();
            }
        }
    }

```

```
    }  
}  
  
private static String readString(String prompt) {  
    while (true) {  
        System.out.print(prompt);  
        String input = scanner.nextLine().trim();  
        if (!input.isEmpty()) {  
            return input;  
        }  
        System.out.println(" [ERROR] Input cannot be empty. Please try  
again.");  
    }  
}
```